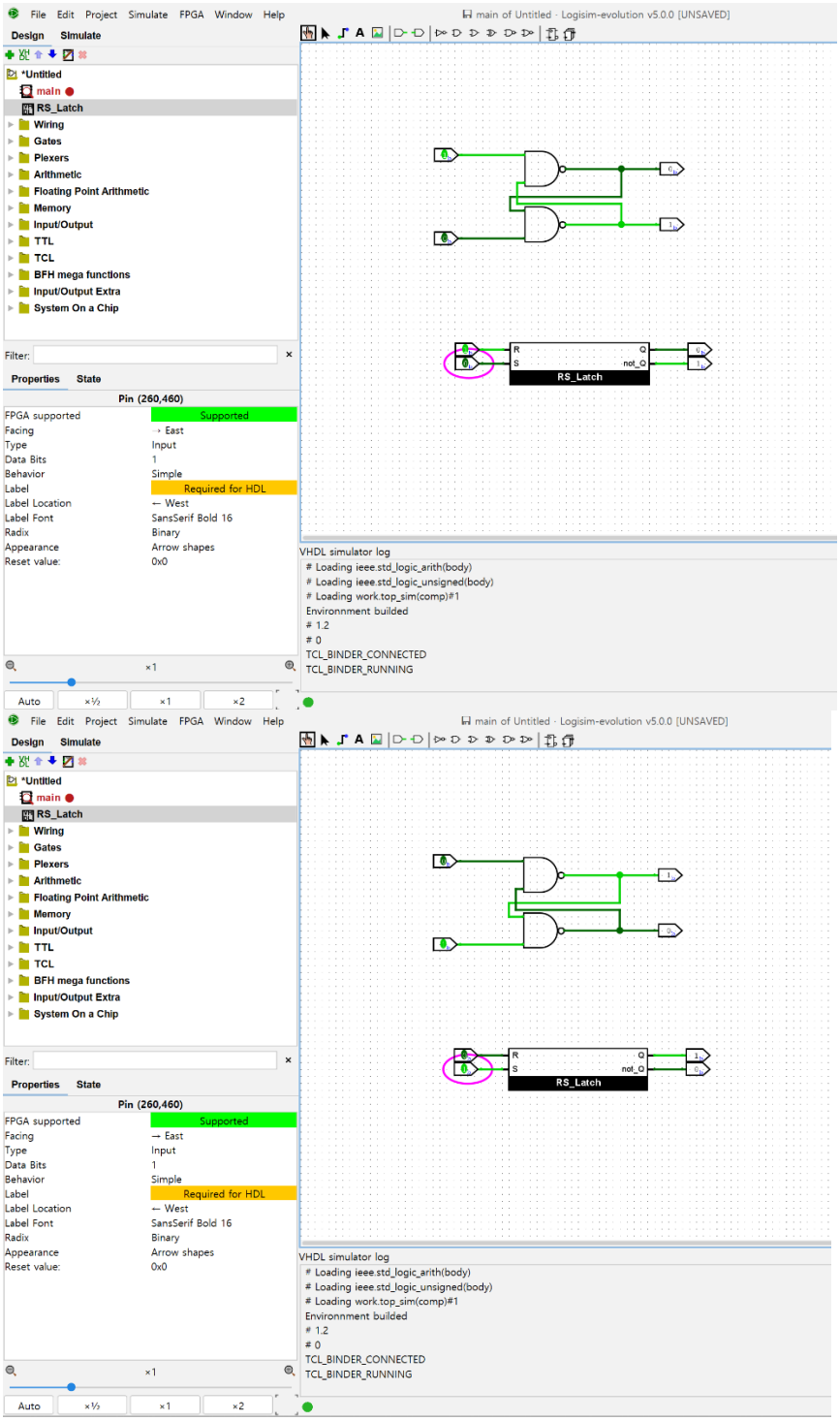
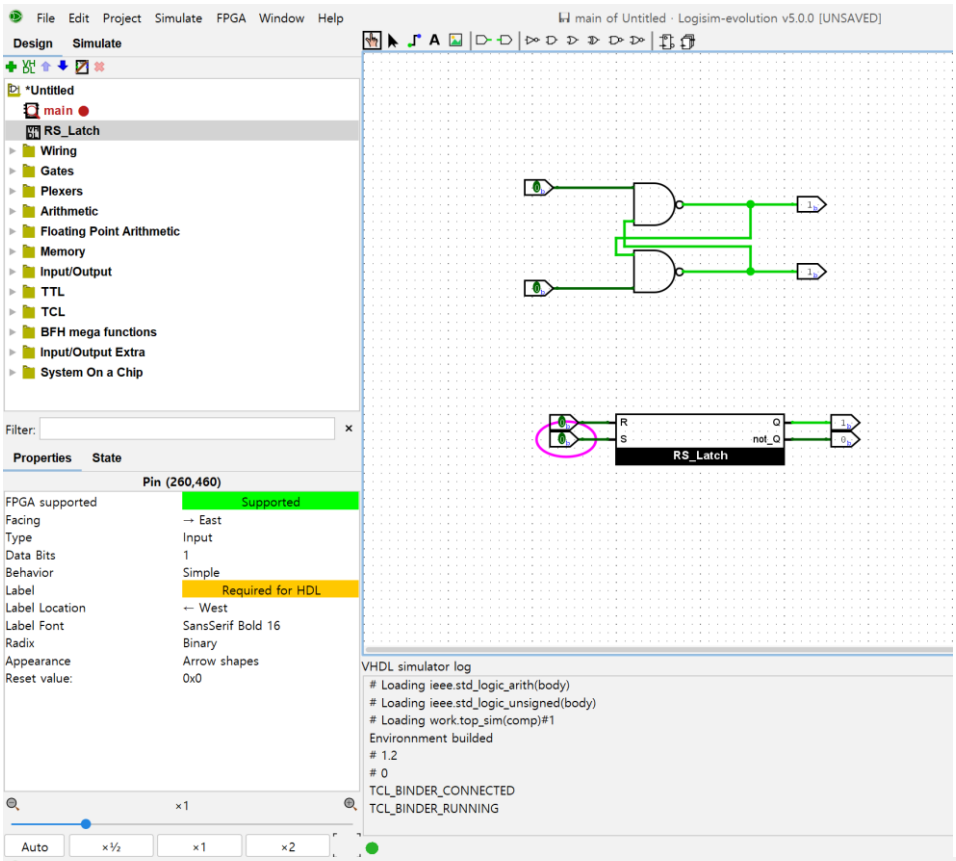
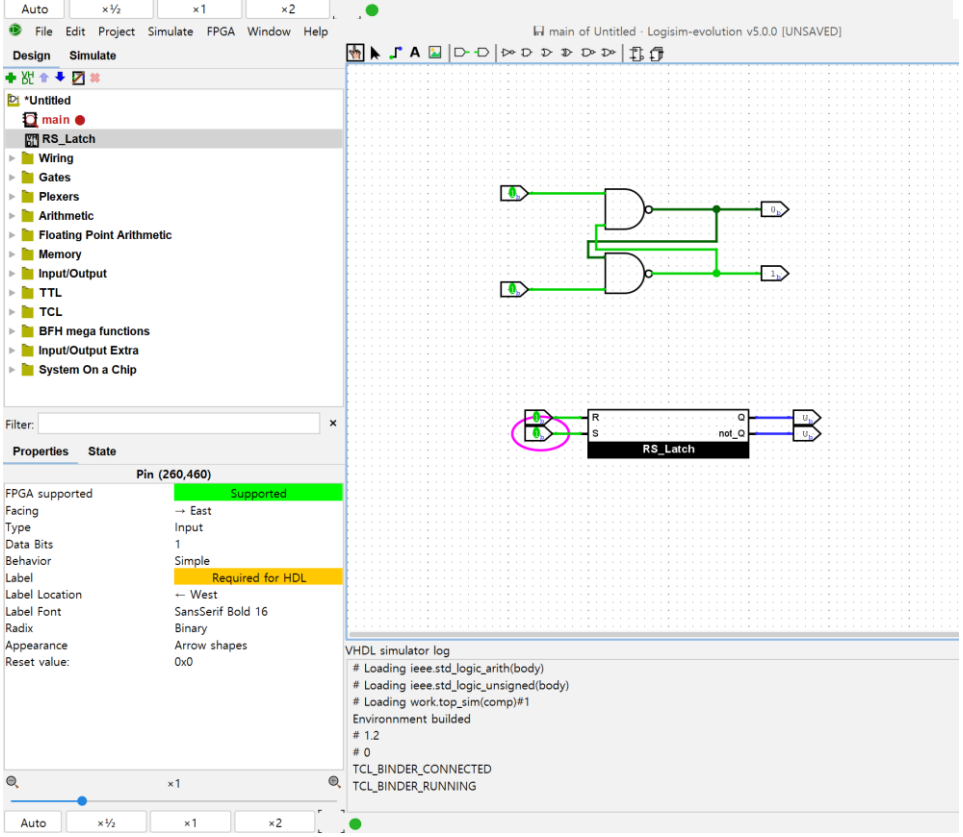
Assignmet3

1) RS Latch

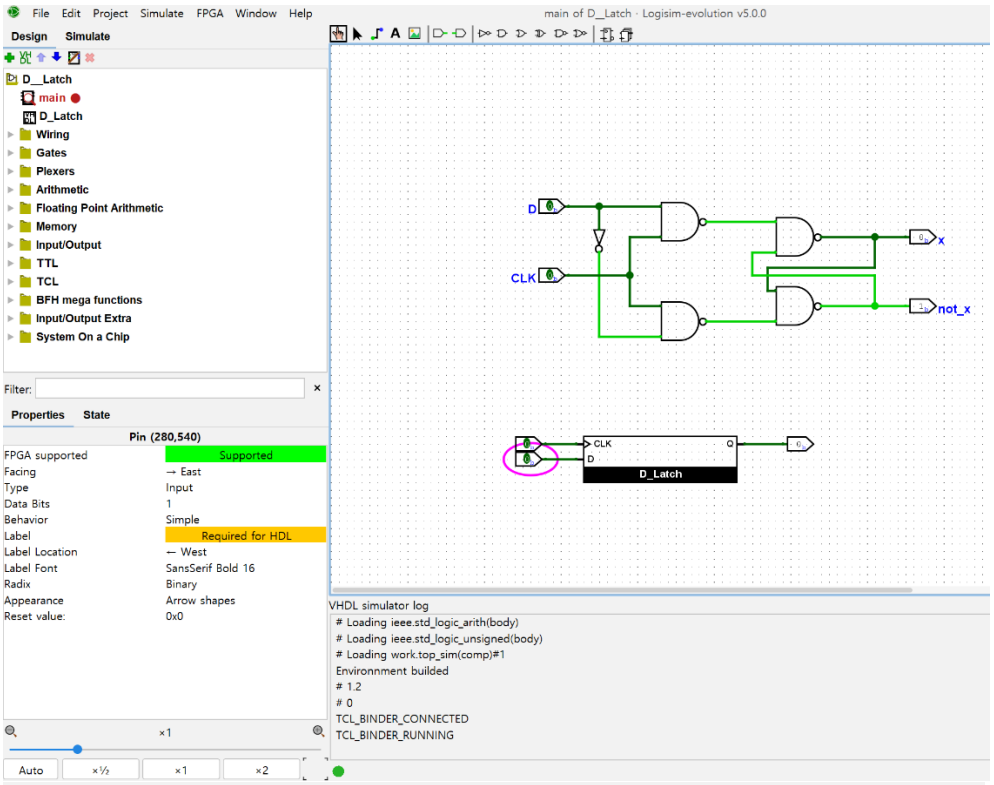


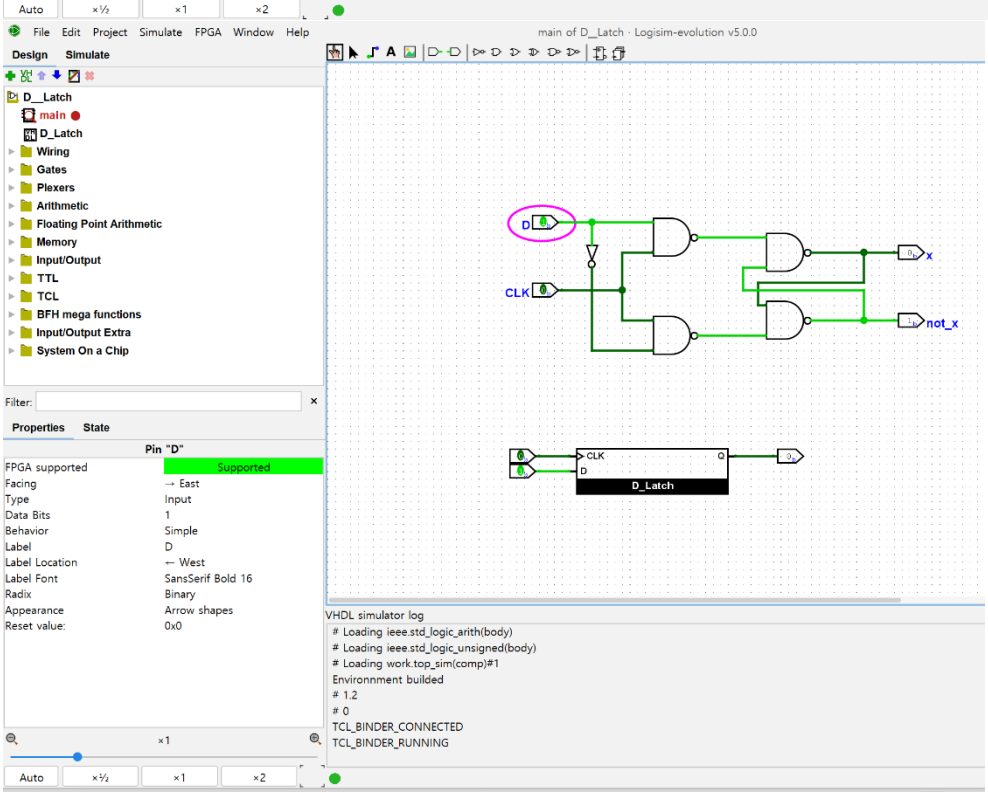

File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
RS_Latch
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (260,460)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
R
S
Q
not_Q
RS_Latch
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
×1
Auto ×½ ×1 ×2

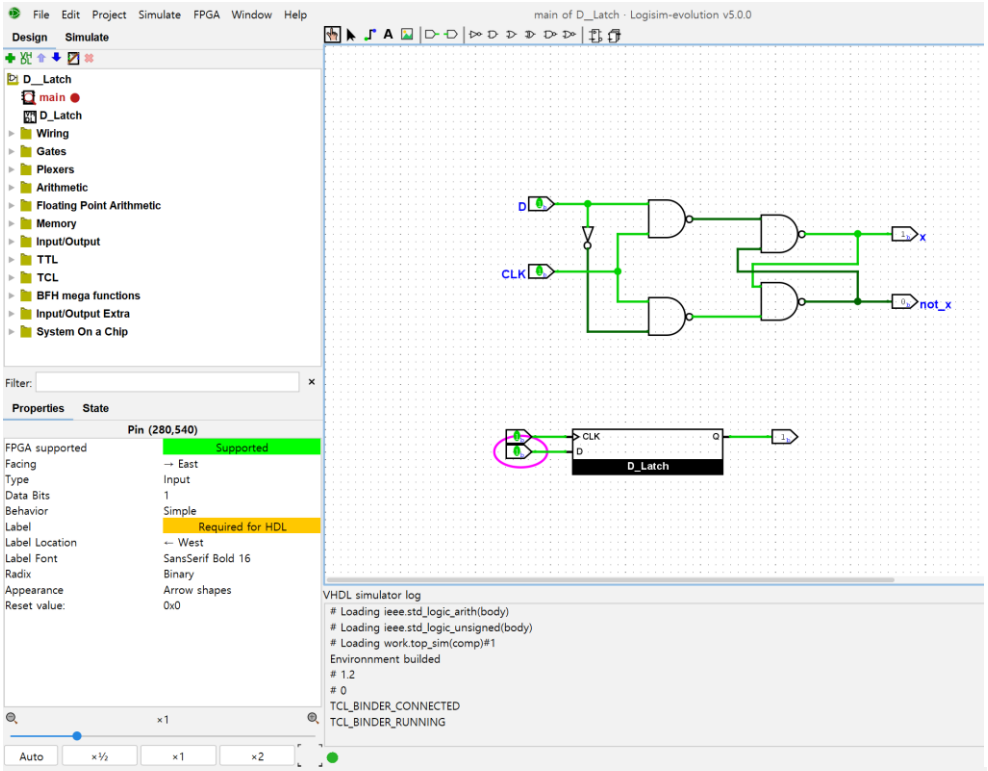

File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
RS_Latch
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (260,460)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
R
S
Q
not_Q
RS_Latch
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
×1
Auto ×½ ×1 ×2

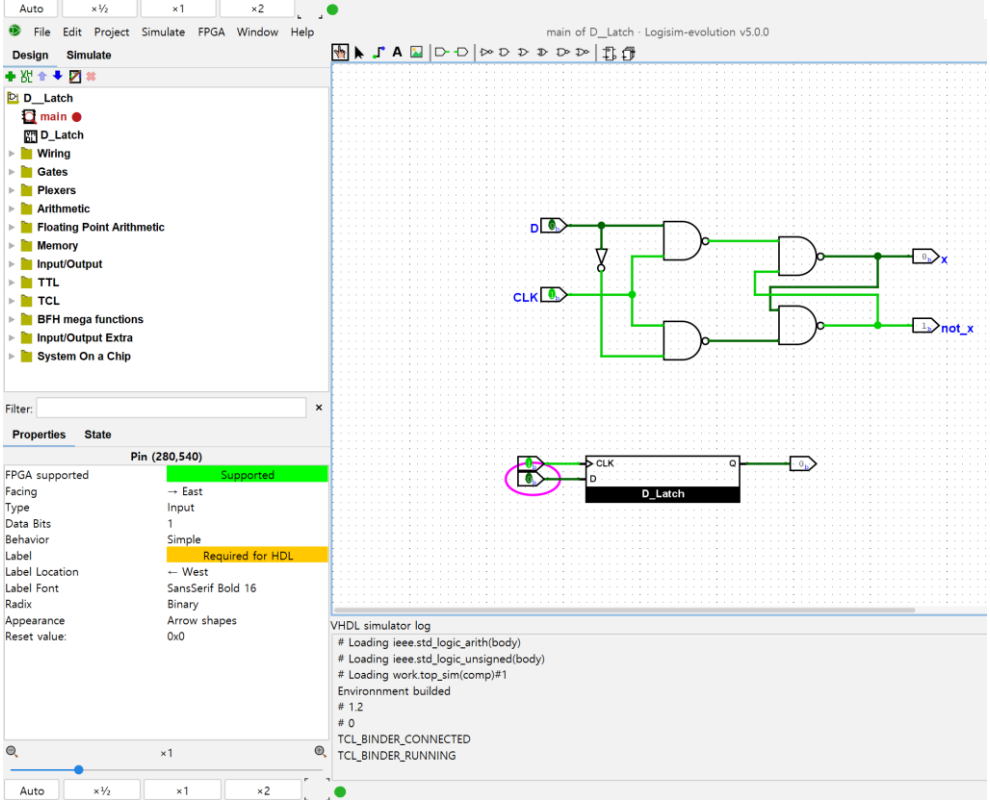
입력값(R, S)에 따라 출력(Q)으로 홀드, 리셋, 셋, 금지 상태를 갖는 회로이다. R=0, S=0일 때 이전값 유지(홀드), R=1일 때 출력 0(리셋), S=1일 때 출력 1(셋), R=1, S=1일 때 금지 상태이다.

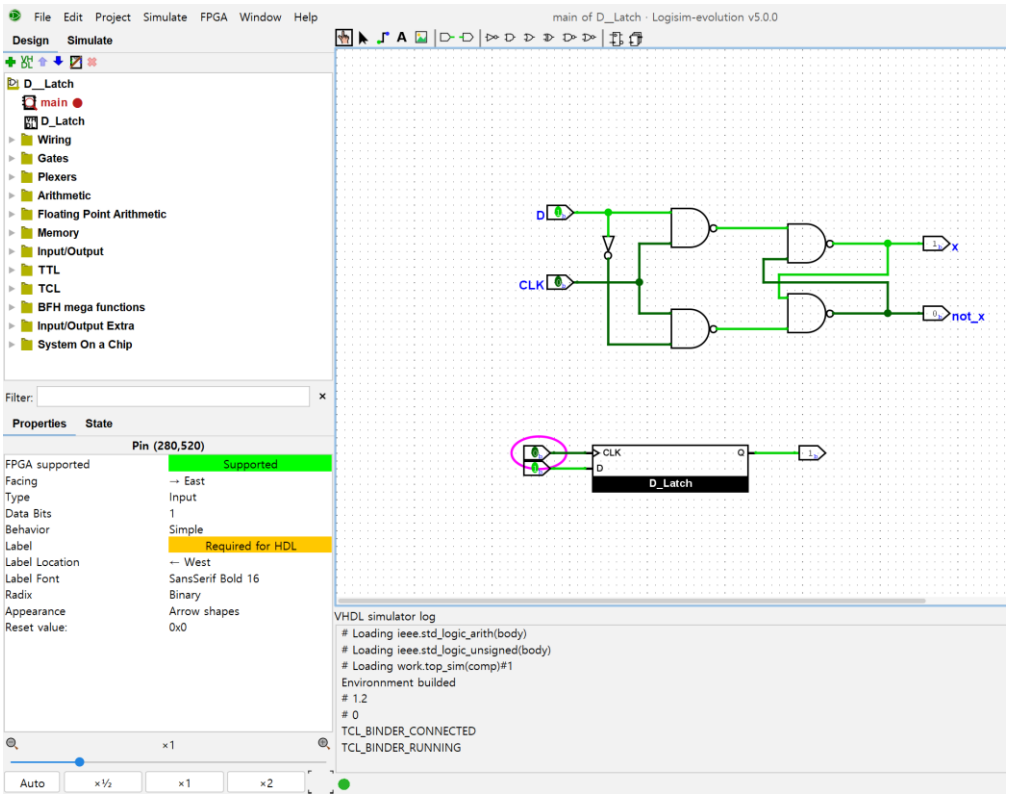
2) D Latch





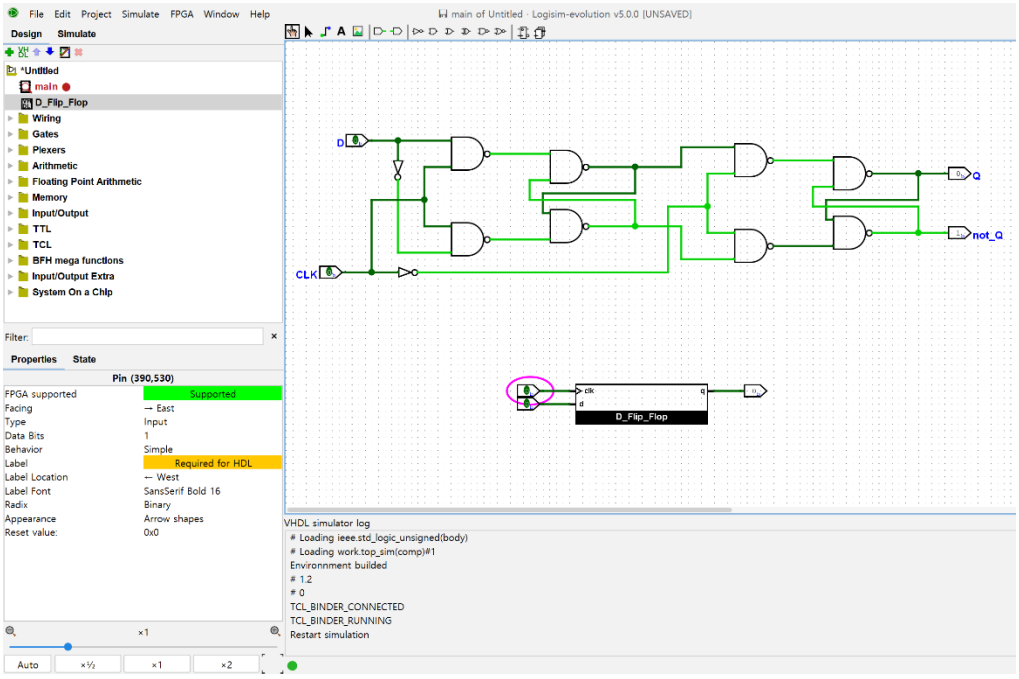

File Edit Project Simulate FPGA Window Help
main of D_Latch · Logisim-evolution v5.0.0
Design Simulate
D_Latch
main
D_Latch
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (280,540)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
×1
Auto ×½ ×1 ×2
D
CLK
x
not_x
CLK
D
Q
D_Latch
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING

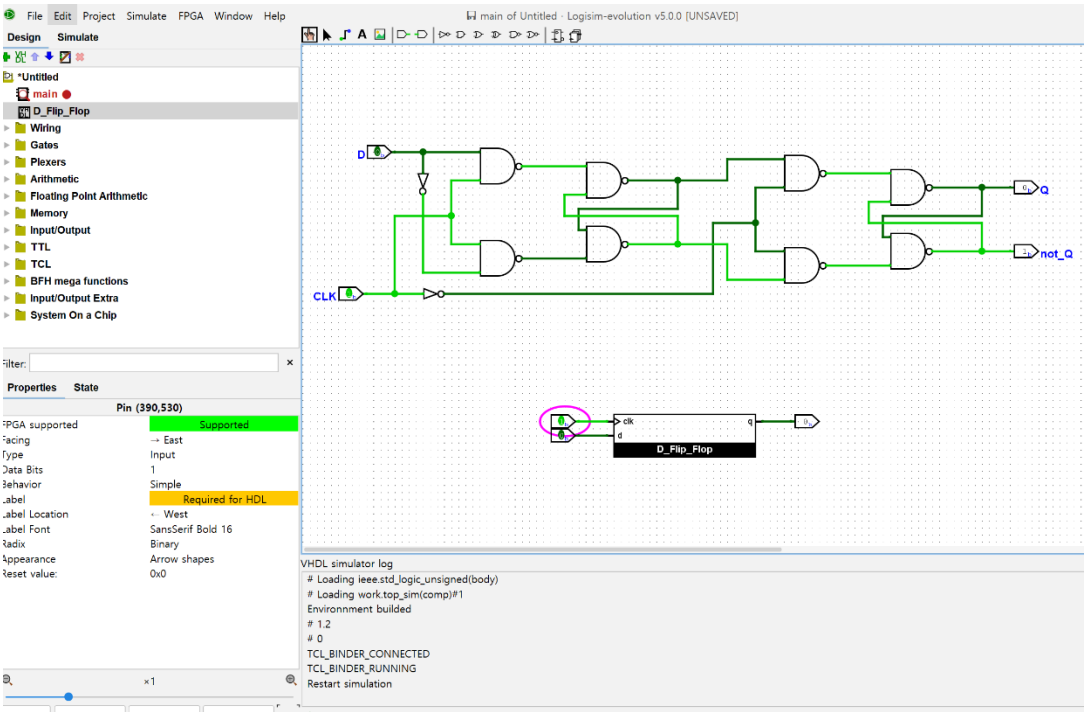

File Edit Project Simulate FPGA Window Help
main of D_Latch · Logisim-evolution v5.0.0
Design Simulate
D_Latch
main
D_Latch
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (280,540)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
×1
Auto ×½ ×1 ×2
D
CLK
x
not_x
CLK
D
Q
D_Latch
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING

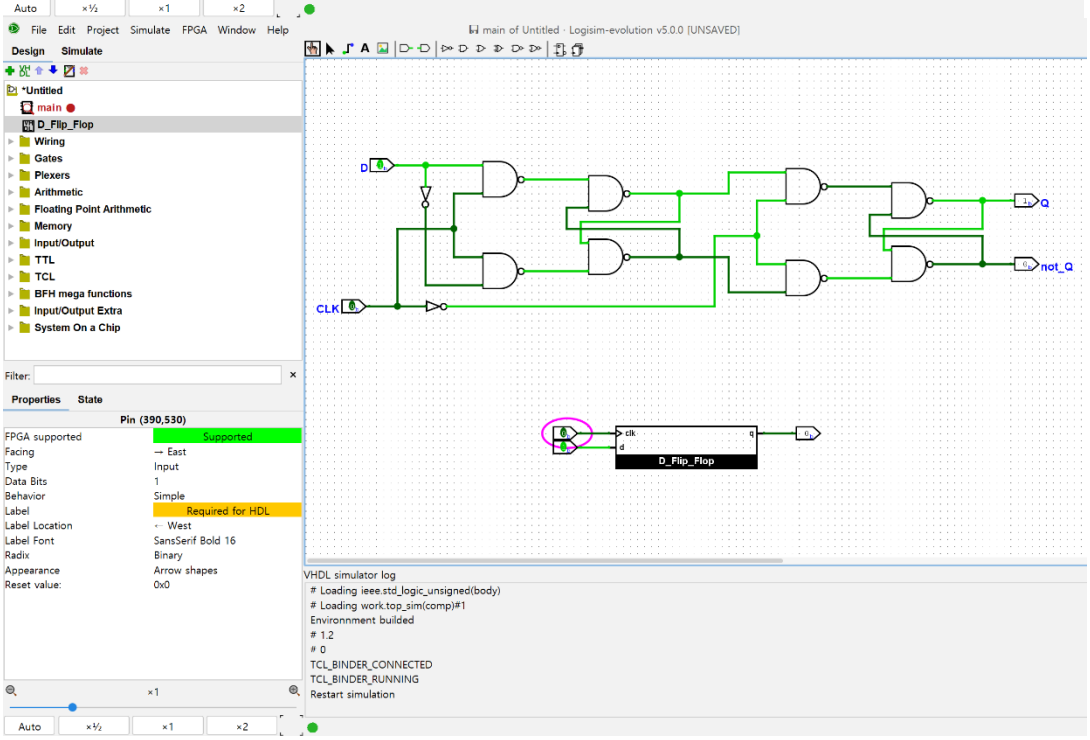


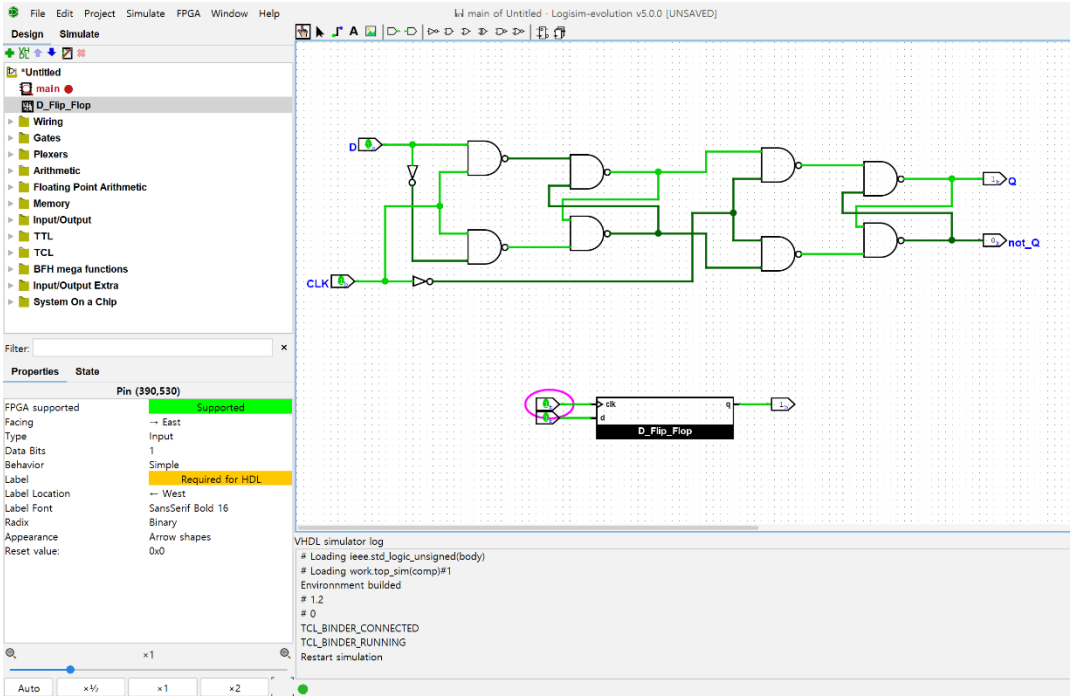
두 입력 중 clk의 값에 따라 출력이 업데이트 되는지 결정되는 회로이다. clk가 0인 경우 d에 어떤 값이 들어오든 출력이 이전 값으로 고정되고, clk가 1인 경우 d에 들어온 값에 따라 출력이 변화한다.

3) D Flip-Flop



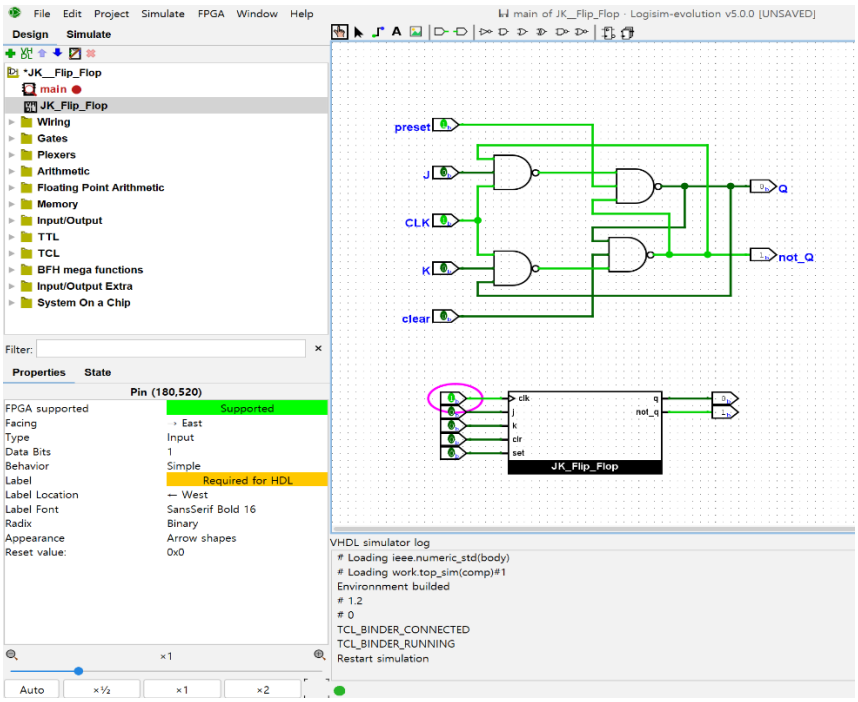

File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
D_Flip_Flop
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Properties State
Pin (390,530)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
D
CLK
Q
not_Q
clk
d
q
D_Flip_Flop
VHDL simulator log
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Restart simulation
Auto ×½ ×1 ×2

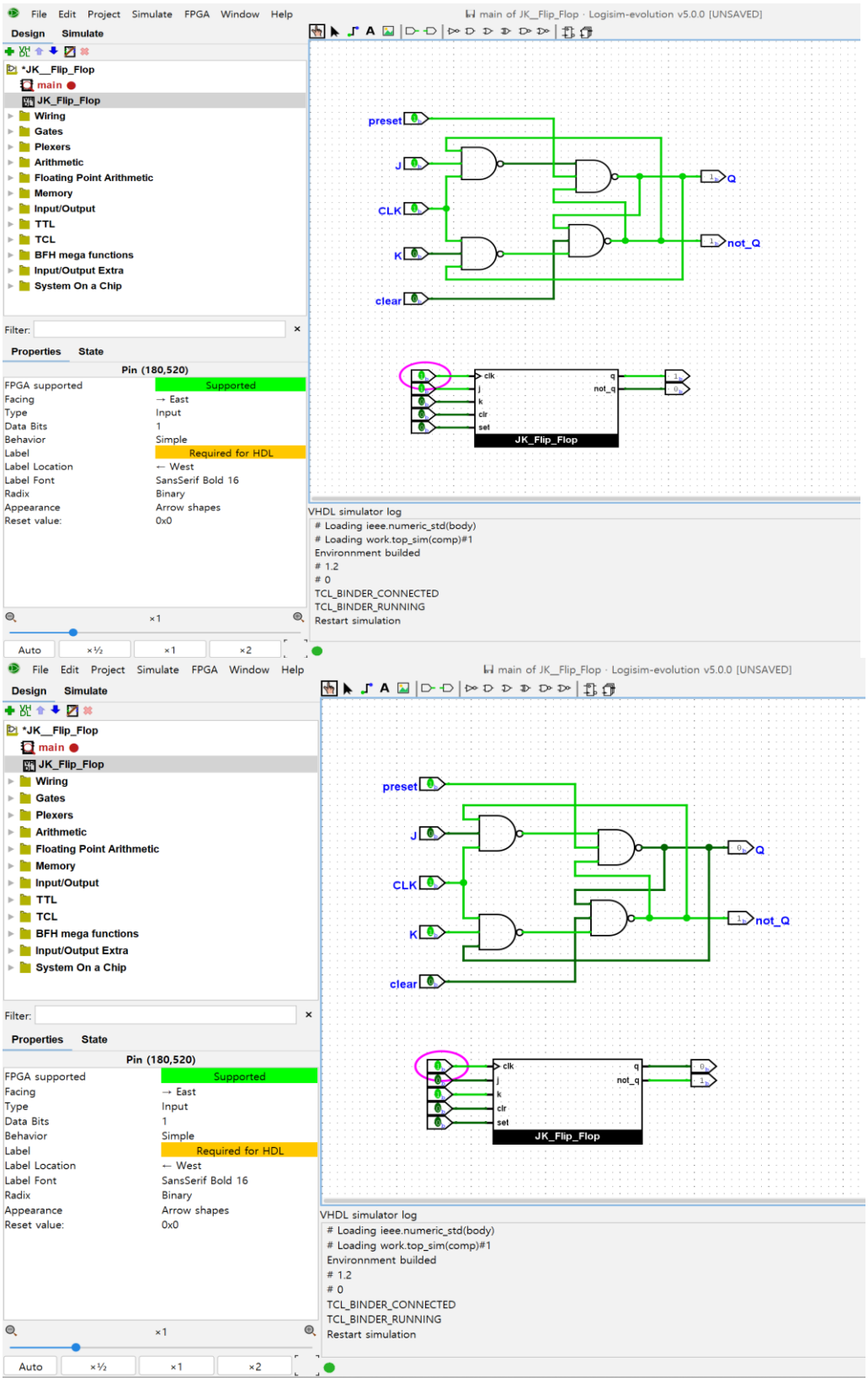

File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
D_Flip_Flop
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Properties State
Pin (390,530)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
D
CLK
Q
not_Q
clk
d
q
D_Flip_Flop
VHDL simulator log
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Restart simulation
Auto ×½ ×1 ×2

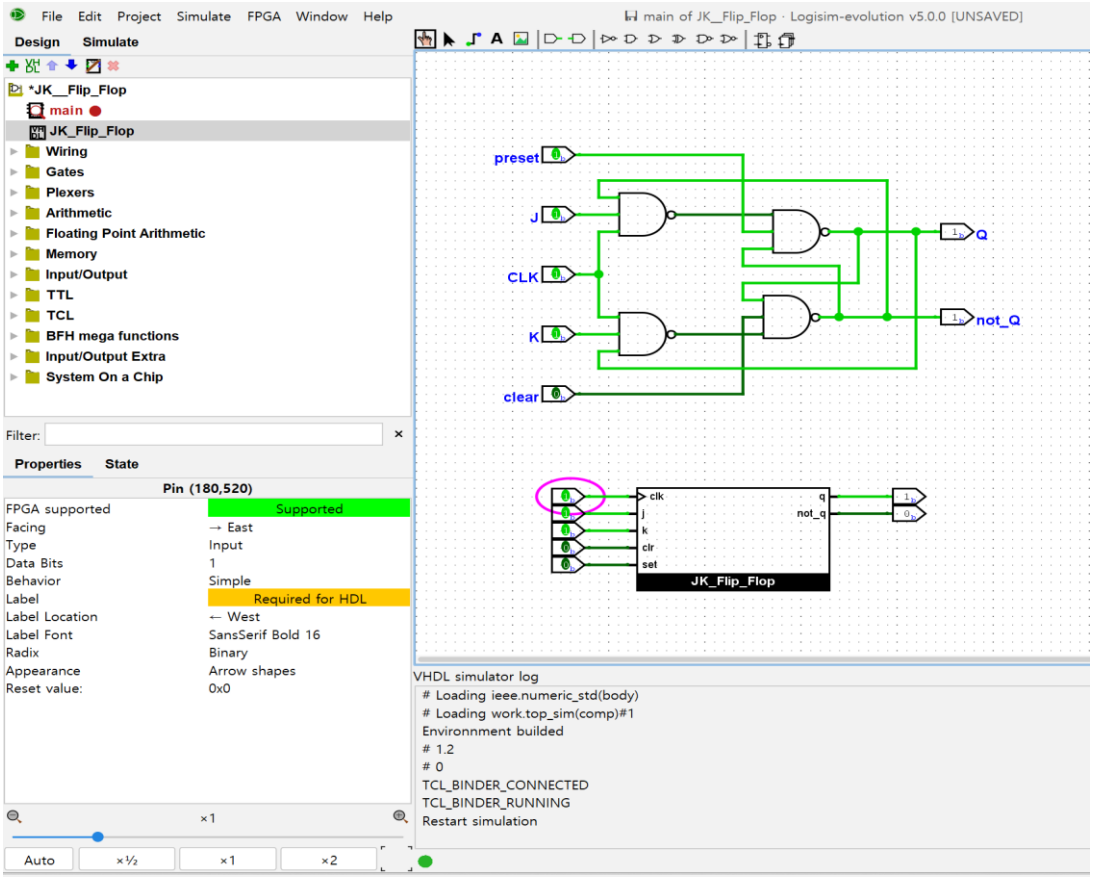


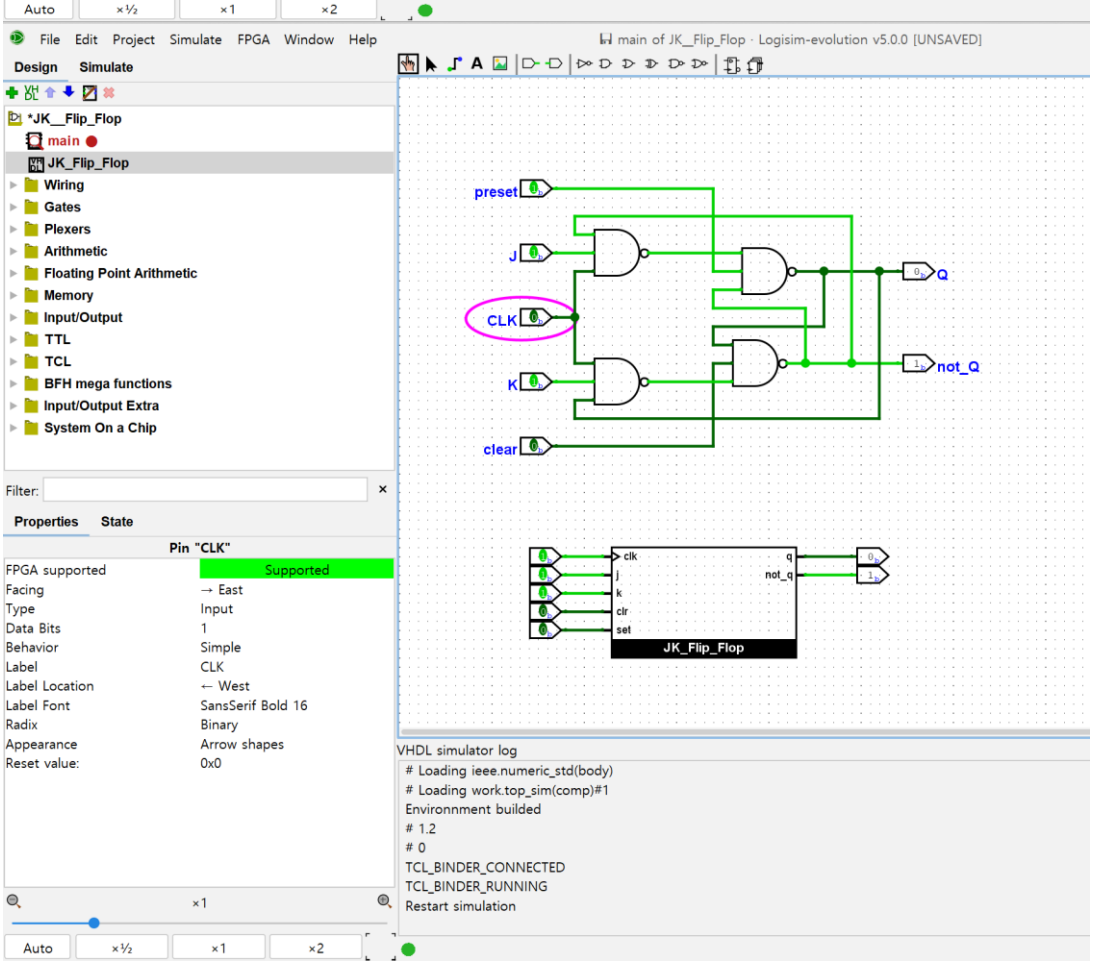
D_Latch 두 개와 inverter로 이루어진 회로로 clk값 변화에 따라 D에 입력된 값의 출력을 지연시키는 회로이다. D에 입력된 값이 변하면 clk가 한 번 변하는 동안 이전 값을 유지하고, clk에 한 번 더 변하면 기억한 변한 D의 값을 출력한다.

4) JK Flip Flop



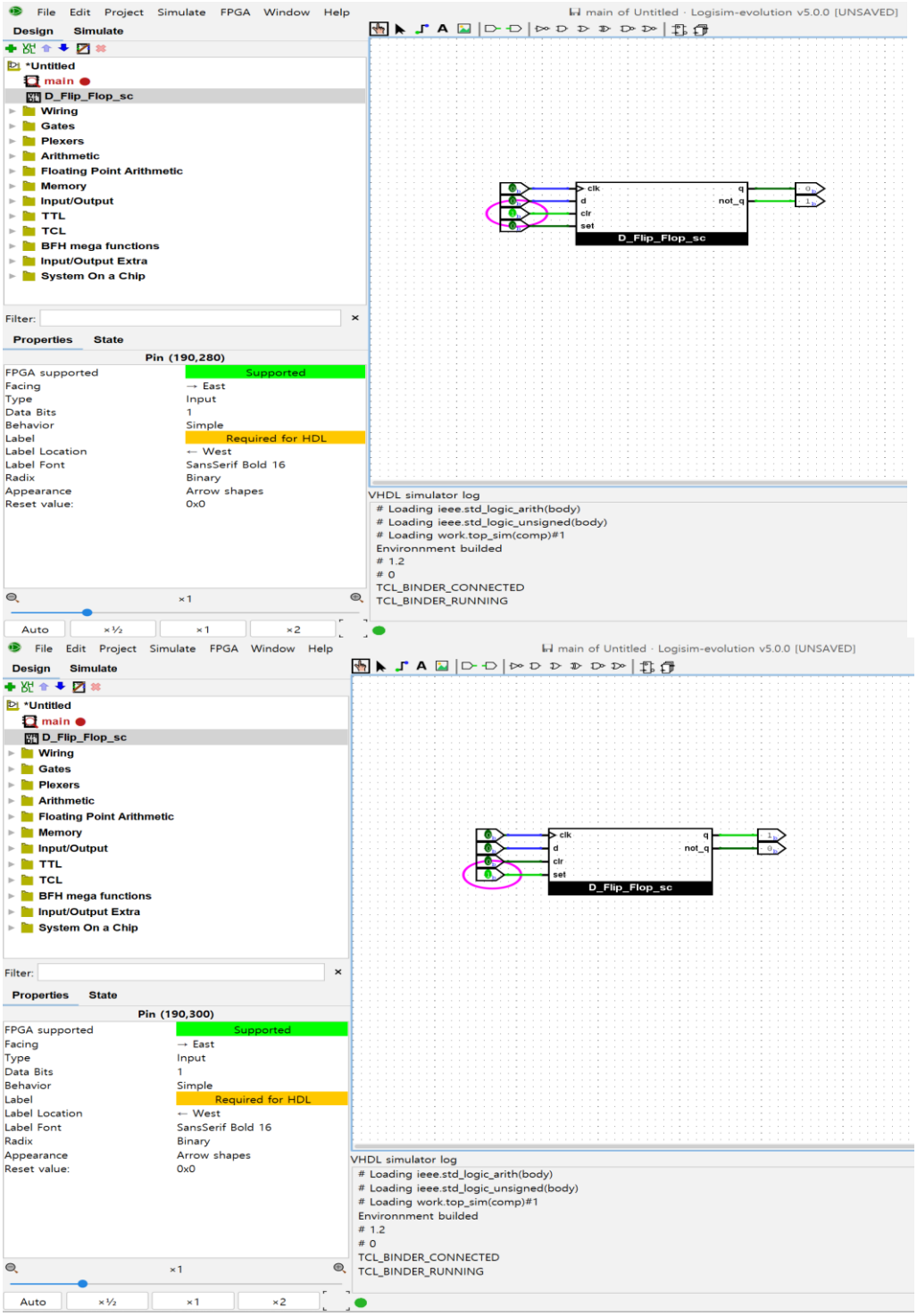

File Edit Project Simulate FPGA Window Help
main of JK_Flip_Flop · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*JK__Flip_Flop
main
JK_Flip_Flop
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (180,520)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
×1
Auto ×½ ×1 ×2
preset
J
CLK
K
clear
Q
not_Q
clk
j
k
clr
set
q
not_q
JK_Flip_Flop
VHDL simulator log
Loading ieee.numeric_std(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Restart simulation

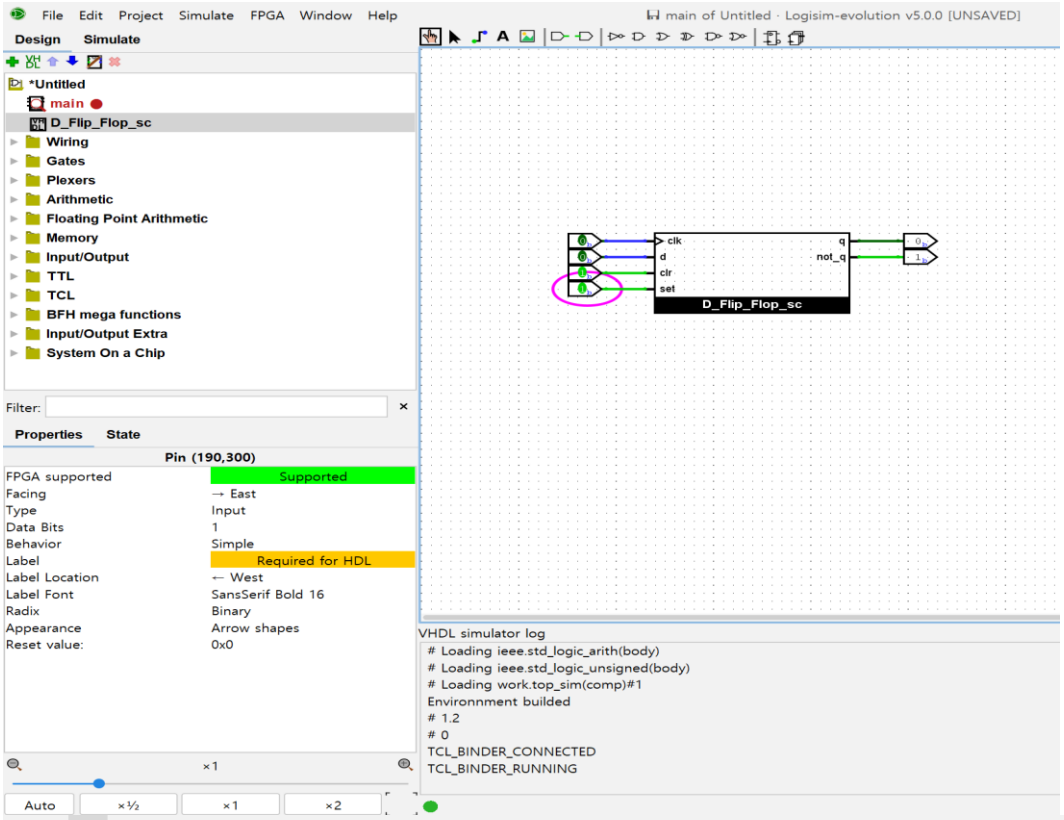

File Edit Project Simulate FPGA Window Help
main of JK_Flip_Flop · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*JK_Flip_Flop
main
JK_Flip_Flop
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (180,520)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
preset
J
CLK
K
clear
Q
not_Q
clk
j
k
clr
set
q
not_q
JK_Flip_Flop
VHDL simulator log
Loading ieee.numeric_std(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Restart simulation
×1
Auto ×½ ×1 ×2

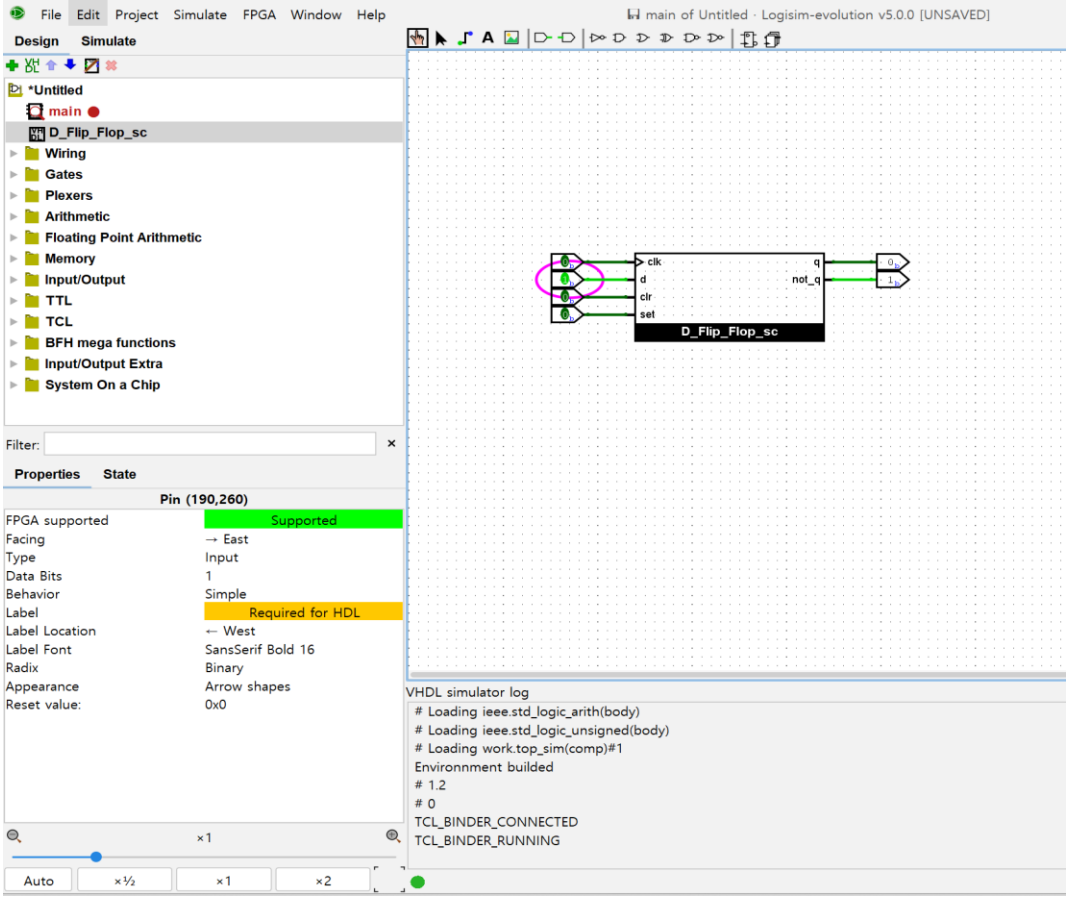

File Edit Project Simulate FPGA Window Help
main of JK_Flip_Flop · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*JK_Flip_Flop
main
JK_Flip_Flop
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin "CLK"
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label CLK
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
preset
J
CLK
K
clear
Q
not_Q
clk
j
k
clr
set
q
not_q
JK_Flip_Flop
VHDL simulator log
Loading ieee.numeric_std(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Restart simulation
×1
Auto ×½ ×1 ×2

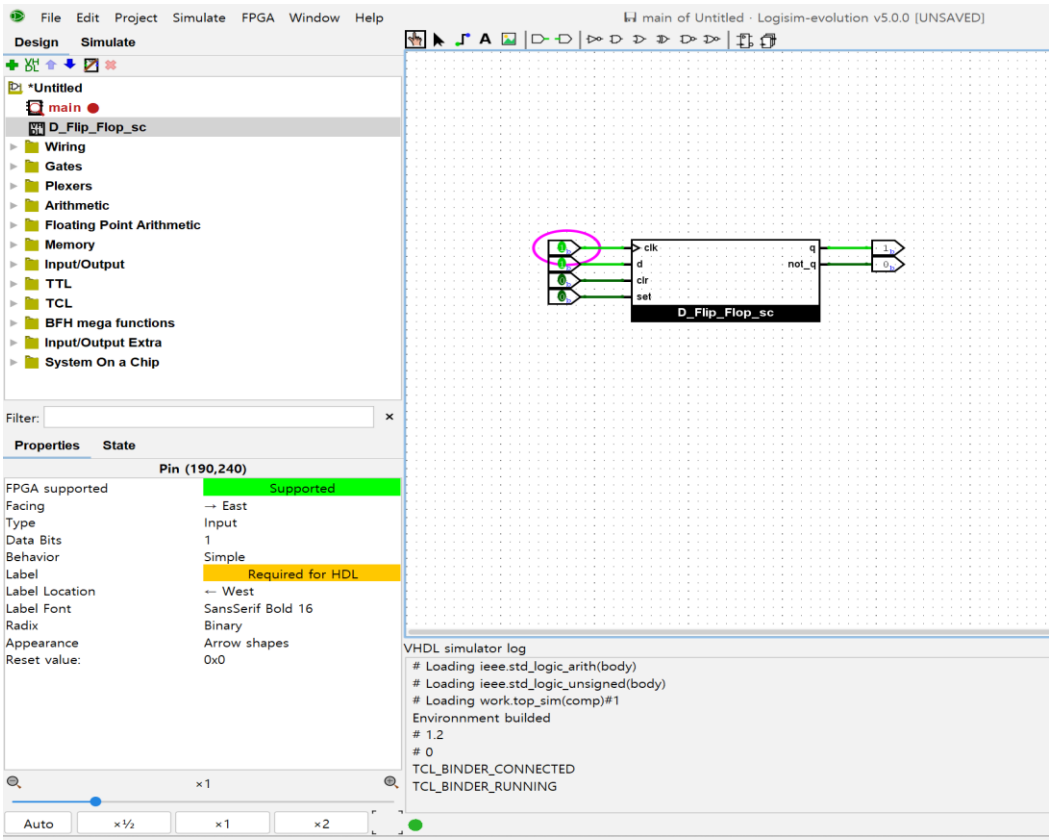
앞서 서술한 RS_Latch와 유사한 기능을 가진 회로이다. RS_Latch와 동일하게 홀드, 셋, 리셋 기능을 가지고 있으며, 추가로 토글 기능이 있다. 앞의 세 기능의 조건과 동작은 RS_Latch와 동일하고, 토글은 J=1, K=1일 때 출력을 현재 출력의 반대로 바꾼다.

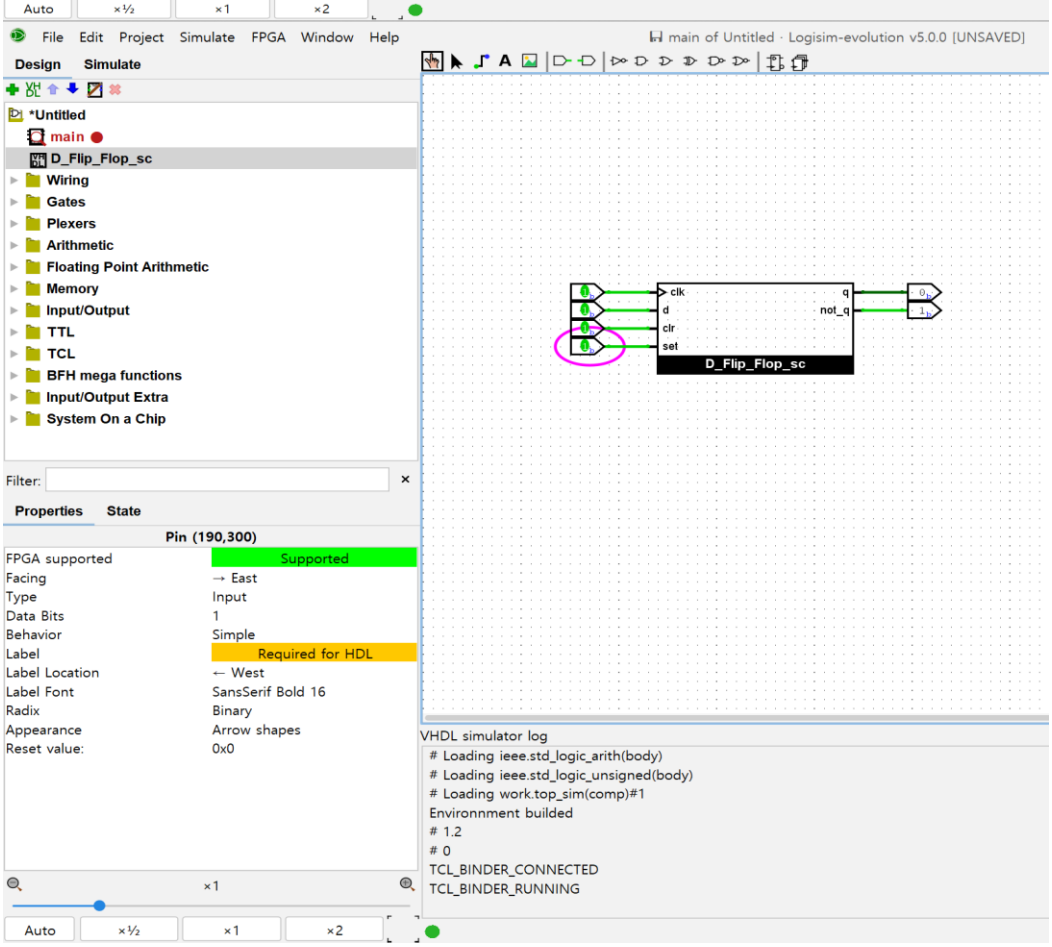
5) D Flip Flop with Asynchronous Set and Clear




File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
D_Flip_Flop_sc
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (190,300)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
clk
d
clr
set
q
not_q
D_Flip_Flop_sc
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Auto ×½ ×1 ×2


File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
D_Flip_Flop_sc
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (190,260)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
clk
d
clr
set
q
not_q
D_Flip_Flop_sc
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Auto ×½ ×1 ×2


File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
D_Flip_Flop_sc
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (190,240)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
clk
d
clr
set
q
not_q
D_Flip_Flop_sc
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
×1
Auto ×½ ×1 ×2


File Edit Project Simulate FPGA Window Help
main of Untitled · Logisim-evolution v5.0.0 [UNSAVED]
Design Simulate
*Untitled
main
D_Flip_Flop_sc
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (190,300)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
clk
d
clr
set
q
not_q
D_Flip_Flop_sc
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
×1
Auto ×½ ×1 ×2

D_Flip_Flop 회로에 clear와 set 기능을 추가한 회로이다. clr에 1이 입력되면 출력이 0으로, set에 1이 입력되면 출력이 1로 고정된다. 이를 이용하여 초기 값을 지정하여 오류를 방지할 수 있다. 이번에 만든 회로는 clr이 set보다 우선으로 적용되는 코드로 작성되어 있어 두 입력에 모두 1이 들어오면 clr의 동작이 우선되어 출력이 0으로 고정된다. 나머지 기능은 기본 D_Flip_Flop과 같다.